

ORF 309 – Homework 4, Q1

Q1(a): Compute $P\{X + Y = k\}$

Setup. Let $X \sim \text{Binom}(n, p)$ and $Y \sim \text{Binom}(m, p)$ be independent random variables.

Probabilistic argument. X counts the number of successes in n independent trials, each with success probability p . Similarly, Y counts successes in m independent trials with the same probability p . Since X and Y are independent, $X + Y$ counts the total number of successes across all $n + m$ independent trials. Therefore:

$$X + Y \sim \text{Binom}(n + m, p)$$

Result.

$$P\{X + Y = k\} = \binom{n + m}{k} p^k (1 - p)^{n + m - k}, \quad k = 0, 1, \dots, n + m$$

Verification via convolution. One can also verify this mechanically using the law of total probability. Since X and Y are independent:

$$\begin{aligned} P\{X + Y = k\} &= \sum_{j=0}^k P\{X = j\} P\{Y = k - j\} \\ &= \sum_{j=0}^k \binom{n}{j} p^j (1 - p)^{n-j} \cdot \binom{m}{k-j} p^{k-j} (1 - p)^{m-(k-j)} \\ &= p^k (1 - p)^{n+m-k} \sum_{j=0}^k \binom{n}{j} \binom{m}{k-j} \end{aligned}$$

By the Vandermonde identity, $\sum_{j=0}^k \binom{n}{j} \binom{m}{k-j} = \binom{n+m}{k}$, which recovers the same answer.

Q1(b): Compute $P\{X = Z - 1\}$

Setup. Let $X \sim \text{Binom}(n, p)$ and $Z \sim \text{Geom}(p)$ be independent. Recall that Z is the number of trials until the first success, so $Z - 1$ is the number of failures before the first success:

$$P\{Z - 1 = k\} = (1 - p)^k \cdot p, \quad k = 0, 1, 2, \dots$$

Calculation. By independence, we sum over all values k that X and $Z - 1$ can share (i.e., $k = 0, 1, \dots, n$):

$$\begin{aligned} P\{X = Z - 1\} &= \sum_{k=0}^n P\{X = k\} \cdot P\{Z - 1 = k\} \\ &= \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} \cdot (1 - p)^k \cdot p \\ &= p (1 - p)^n \sum_{k=0}^n \binom{n}{k} p^k \end{aligned}$$

By the binomial theorem, $\sum_{k=0}^n \binom{n}{k} p^k \cdot 1^{n-k} = (1 + p)^n$. Therefore:

$$\begin{aligned} P\{X = Z - 1\} &= p (1 - p)^n (1 + p)^n \\ &= p [(1 - p)(1 + p)]^n \end{aligned}$$

Result.

$$P\{X = Z - 1\} = p (1 - p^2)^n$$

Appendix: The Binomial Theorem

Statement. For any a , b , and positive integer n :

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

Proof (combinatorial)

$(a + b)^n$ means multiplying n copies of $(a + b)$ together:

$$\underbrace{(a + b)(a + b)(a + b) \cdots (a + b)}_{n \text{ factors}}$$

When you expand this product, each term comes from choosing either a or b from each of the n factors. A term with $a^k b^{n-k}$ arises when you pick a from exactly k factors and b from the remaining $n - k$ factors.

How many ways can you pick a from exactly k out of n factors? That is $\binom{n}{k}$ (“ n choose k ”).

So the coefficient of $a^k b^{n-k}$ is $\binom{n}{k}$, and summing over all possible values of k gives:

$$(a + b)^n = \binom{n}{0} a^0 b^n + \binom{n}{1} a^1 b^{n-1} + \cdots + \binom{n}{n} a^n b^0 = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad \square$$

Concrete numerical example

Let's verify with $a = 1$, $b = 2$, $n = 3$. We know $(1 + 2)^3 = 3^3 = 27$.

Now expand using the theorem:

$$\begin{aligned} (1 + 2)^3 &= \binom{3}{0} 1^0 \cdot 2^3 + \binom{3}{1} 1^1 \cdot 2^2 + \binom{3}{2} 1^2 \cdot 2^1 + \binom{3}{3} 1^3 \cdot 2^0 \\ &= 1 \cdot 8 + 3 \cdot 4 + 3 \cdot 2 + 1 \cdot 1 \\ &= 8 + 12 + 6 + 1 = 27 \quad \checkmark \end{aligned}$$

How it was used in Q1(b)

We had the sum:

$$\sum_{k=0}^n \binom{n}{k} p^k$$

This matches the binomial theorem with $a = p$ and $b = 1$:

$$\sum_{k=0}^n \binom{n}{k} p^k \cdot 1^{n-k} = (p + 1)^n = (1 + p)^n$$

This is the step that collapsed the sum into the closed-form expression.